

Garth Mortensen

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Production Data Science

Full stack production-grade data scientist with strong software engineering background. Love for creative destruction: building pipelines, reducing complexity, and designing clean, maintainable architectures. Let's talk.

Example Proof of Concepts/

Prism

Prism is a modernized Risk Adjustment platform which replaces opaque “black box” legacy tools with a transparent, code-first architecture. It treats Risk Adjustment as a software engineering problem, ensuring usability, accuracy, auditability, and rapid scenario planning. Analysts parametrize model parameters, and guard-railed **AI agents interpret** and process downstream results.

SpilloverLab.com

Time series analysis platform which allows users to experiment with financial market spillover modeling. Delivers simple-English interpretations, dashboards, [REST](#) and [GraphQL](#) APIs for advanced users. Constructed with microservices, deployed on cloud with CI/CD [auto-documentation](#), [tested](#) and [unit testing](#).

Professional Experience

Federal Reserve Bank of Minneapolis

Sr. Production Data Scientist, National Stress Testing Program. *Minneapolis, MN - 06/2022 – Present*

- AI: Delivered RAG (Retrieval-Augmented Generation) system for Stress Testing program, enabling retrieval of relevant documentation to support analysis. Prototyped agentic workflows to automate execution, information retrieval, model configuration and decision support.
- Hired to productionalize credit risk models; expanded role to assume ownership of orchestration, ETL, and production infrastructure supporting the stress testing program. -Productionalized mortgage and credit card risk models and performed ad hoc analyses consumed by the Board of Governors.

- Owned orchestration platform responsible for congressionally mandated Dodd-Frank Stress Testing workflows, ensuring reliable execution of complex data pipelines.
- Independently designed and delivered a modern ETL framework, reducing data loading times from two weeks to ten minutes.
- Delivered six infrastructure releases across Python, SQL, R, and Bash, improving reliability and usability to support experimentation at scale.
- Led migration of an enterprise-scale production codebase across operating systems, language versions, and platforms.
- Served 120+ quantitative analysts as a technical resource for Python, databases, Linux, and Git.
- Acted as Production's liaison and implementation partner to frontend, database, systems, and DevOps teams.
- Mentored analysts into contributing to the production codebase, improving team capability and knowledge sharing.

University of Minnesota Coding Bootcamp

Fintech Instructor. [Student Review](#). *Saint Paul, MN - 09/2021 – 04/2023*

- Taught programming to professionals ranging from sysadmins to finance directors.
- Guided students in coding up ML models, blockchain implementations, Solidity smart contracts.
- Across 270+ lecture hours, scored 90% across all rating categories (engagement, clarity, knowledge).

Medica

Sr. Risk Adjustment Analyst. *Minnetonka, MN - 11/2019 – 06/2022*

- Implemented ML models to predict member health scores.
- Independently discovered how to tie-out government reporting database with data warehouse.
- Configured bare-metal server to run automated OCR process to ETL .pdf into database, as ML model input.
- In zero-sum marketplace, consistently outperformed leading health insurance companies, boosting revenue.
- High-impact contributions recognized with a 12% discretionary merit increase.

United Health Group R&D

Developer/Data Scientist, contract. *Minneapolis, MN - 01/2019 – 11/2019*

- Replicated academic research on internal data to uncover actionable insights presented to the CEO.

- Built end-to-end data pipelines: API ingestion, transformation, validation, and visualization.
- Contributed to the design and development of a next-gen big data platform for health research.

Munich Reinsurance

Sr. Data Analyst. *Minneapolis, MN - 05/2017 – 12/2018*

- Transformed and ingested unstructured claims data into a structured 7M+ record star-schema database.
- Analyzed the resulting dataset to surface insights, including reports for top diagnoses by cost and prevalence, and ML models to predict claims.
- Built and deployed custom tools with clean, user-friendly UI frontends.

Decyphed

Entrepreneurial NLP Dev. *Paris, France - 05/2015 – 01/2016*

- Programmed text analysis tools specialized in language learning applications.

Saylor Consulting

Industry Cost Consultant, contract. Recommendation available. *San Francisco, CA - 12/2013 – 07/2014*

- Analyzed industry factors influencing construction industry costs (material/labor) and forecast prices.
- Overhauled cost benchmarking using early industry indicators, resulting in algorithmic, reproducible cost setting.
- Delivered data-driven market analysis reports and product presentations to Federal client.

GE Capital REIM

Investor Relations. *Paris, France - 07/2012 – 12/2012*

- Oversaw financial database, competitor and industry developments, financial metrics.
- Interpreted market fundamentals for big-ticket investors seeking to purchase entire real estate fund portfolio.
- All \$25+ billion in assets sold off to Blackstone and other financial institutions.

Xinhua News Agency

Financial News. *Beijing, China - 02/2009 – 07/2011*

- Led a 15+ person international team to launch a new economics news show, working 6+ months without PTO to meet a high-stakes deadline.
- Developed a demand driven airtime allocation algorithm based on API driven topic popularity; the method was later leaked to CNN International and appeared on their daily broadcast lineup.
- The program successfully sold content to major global news networks including CNN and Nikkei.

Caijing Magazine

Economic Research Assistant. *Beijing, China - 01/2008 – 09/2008*

- Published 15+ print stories, commodity reports, and analyses.
- Administered and redesigned website. Introduced self made podcasts, video stories, user comments, and other Web 2 technologies, doubling online readership.
- Managed the internship program, allocated work based on strengths and interests.

College, high school jobs

Minneapolis, Minnesota. 1/2001 – 12/2007

- College: Owner of brick-and-mortar studio providing website design, SEO and graphic design services.
- High school: Employee of small business providing graphic design, SEO and offsite hosting/storage services.

Formal Education

St. Thomas University

St. Paul, USA. 2023

- Double MSc in Software Engineering and Data Science
- Double MSc Certifications in AI and Big Data
- GPA 3.9

EDHEC Business School

Nice, France. 2012

- MSc Finance

Hamline University

St. Paul, USA. 2007

- BA Economics

Skills

Tech: Python/R, git, databases (sql/nosql), ETL, linux (shell, RHEL), HPCs, permissions, APIs, cloud (AWS/Google), devops CI/CD, containers, integration, productionalization

Certifications: MS Software Development, MS Databases, MS IT Networking, MS IT Security, MS HTML/CSS, AWS Cloud Practitioner, Graph SQL, Google Analytics IQ, Excel/VBA, Bloomberg Terminal.

Languages: English, French and Mandarin Chinese (diminishing CEFR B2).